

RecycleSense AI

Intelligent Recycling Assistant Using
Artificial Intelligence
Internship Project - IMB AI for Sustainability
Virtual Internship

Prepared By:

Jeeva A

B.E. Computer Science and Engineering

Easwari Engineering College

Technology Stack:

Python, Streamlit, TensorFlow,

MobileNetV2, Pandas

Sustainable Development Goals:

SDG 11 • SDG 12 • SDG 13

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PROBLEM STATEMENT



- Current waste segregation methods are inefficient and heavily dependent on manual effort.
- Many recyclable materials are disposed of incorrectly due to a lack of awareness and proper guidance.
- Manual waste classification is time-consuming, inconsistent, and difficult to scale.
- Improper disposal of plastic, organic, and electronic waste contributes to environmental pollution and resource loss.
- There is a growing need for an intelligent system that can automatically identify waste and provide accurate recycling

PROPOSED SOLUTION – RECYCLESense AI

User Uploads Waste Image



**AI Object Detection
(MobileNetV2)**



**Waste Category
Mapping**



**Environmental
Impact Analysis**



Recycling Guidance



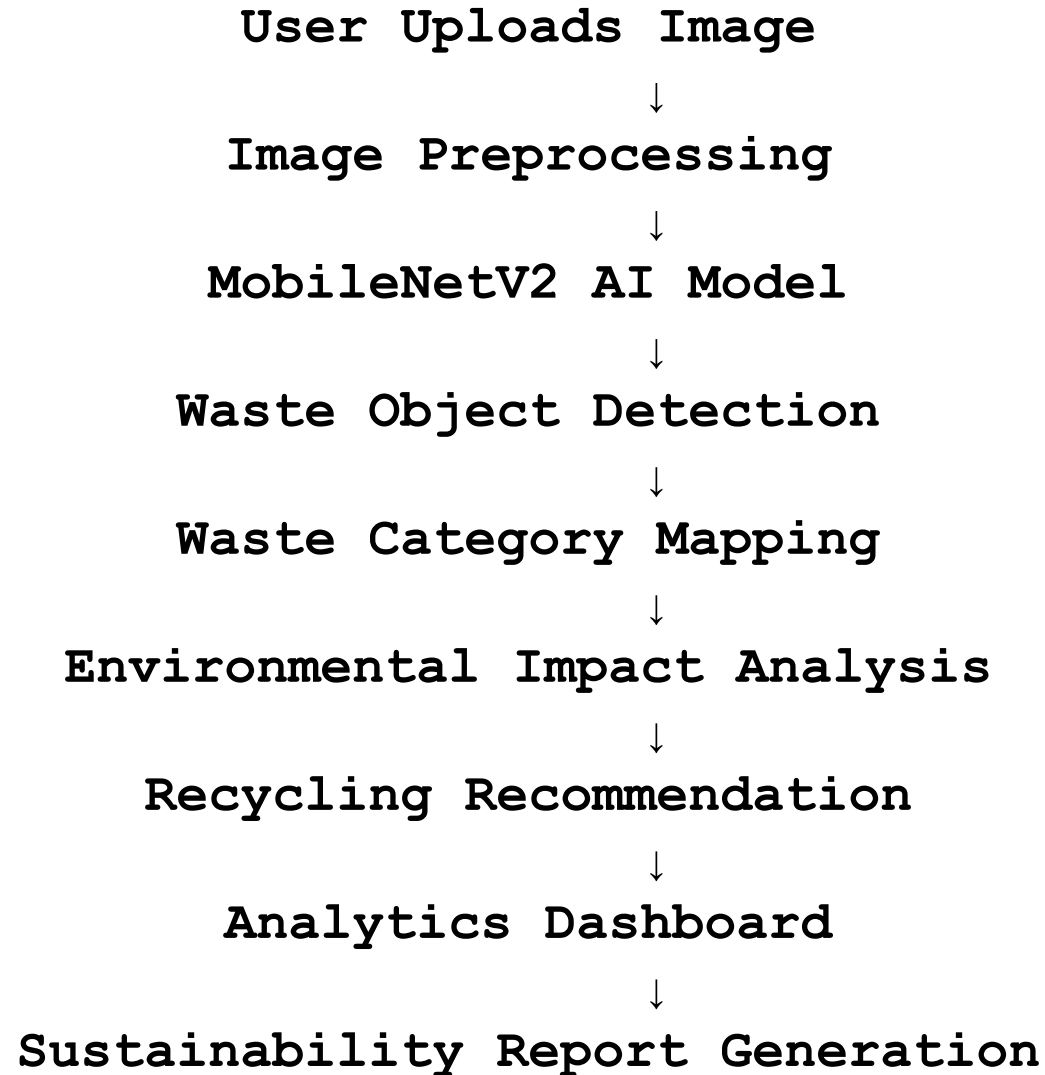
**Analytics Dashboard
&
Sustainability
Report**

- RecycleSense AI is an intelligent waste management platform that uses Artificial Intelligence to identify waste materials from images.
- The system automatically classifies waste into categories such as Plastic, Organic, and E-Waste.
- Users receive disposal recommendations, recycling tips, environmental impact information, and sustainability insights.
- The platform also maintains historical records and generates sustainability reports for analysis.

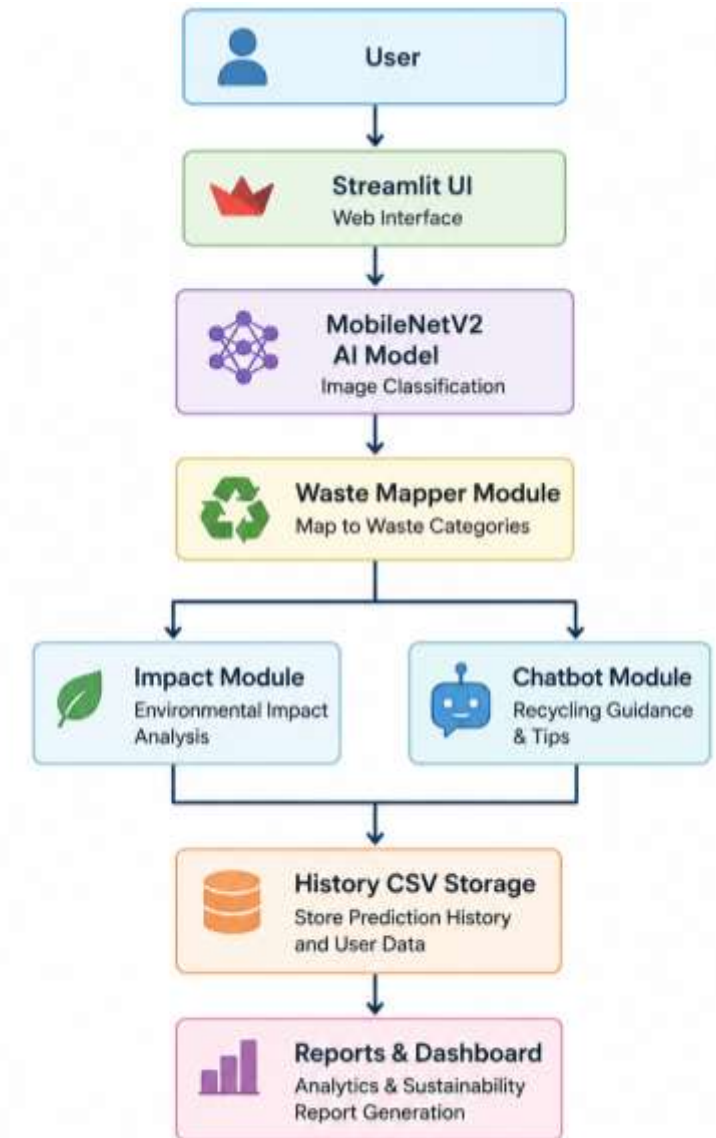
TECHNOLOGY STACK

Component	Technology Used
Programming Language	Python
Frontend	Streamlit
AI Model	MobileNetV2
Machine Learning Library	TensorFlow
Data Processing	Pandas
Visualization	Matplotlib
Data Storage	CSV
Version Control	Git & GitHub

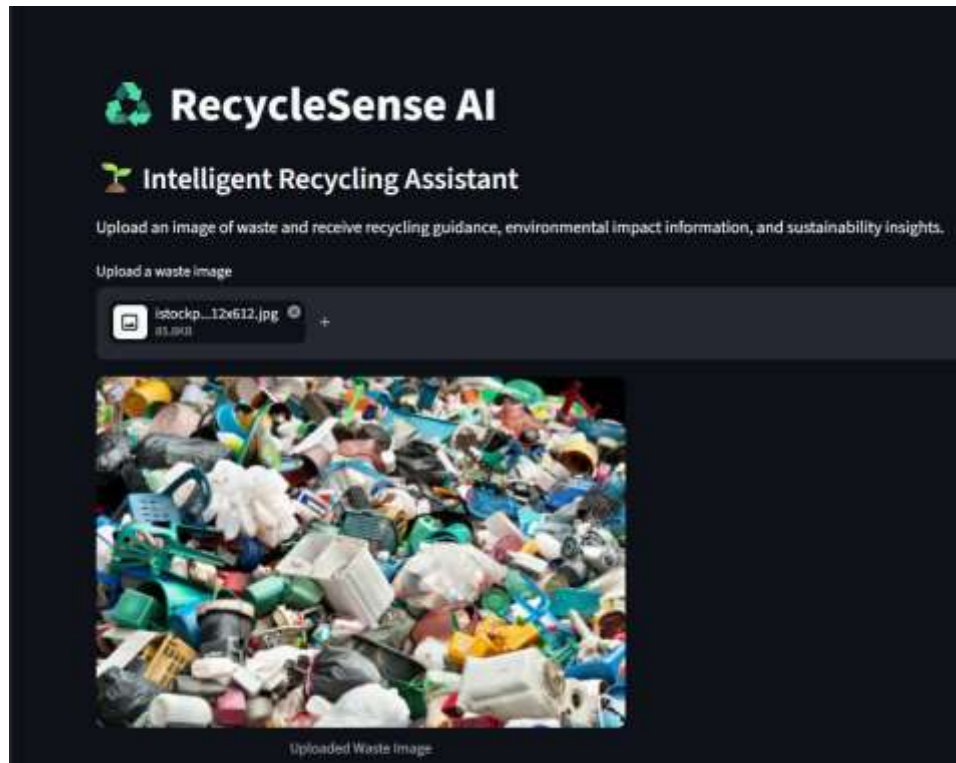
SYSTEM WORKFLOW



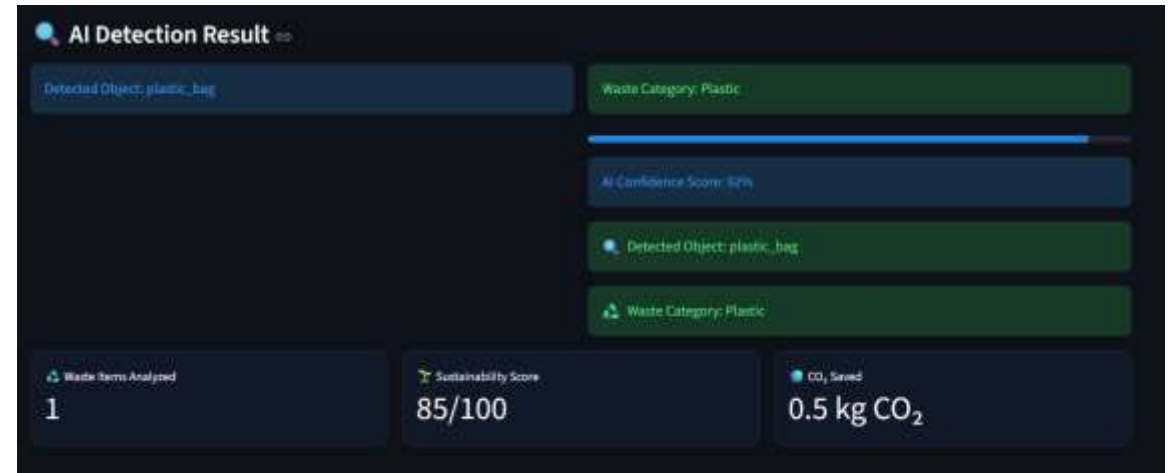
APPLICATION ARCHITECTURE



APPLICATION DEMONSTRATION



Waste Image Upload Interface



AI Waste Detection & Classification



Analytics Dashboard & Detection History

KEY FEATURES

AI Waste Classification

- Uses MobileNetV2 deep learning model
- Identifies waste objects from uploaded images
- Provides fast and accurate predictions

Waste Category Mapping

- Maps detected objects into waste categories
- Categories include:
 - Plastic
 - Organic
 - E-Waste

Environmental Impact Analysis

- Calculates environmental benefits
- Displays CO₂ reduction information
- Promotes sustainability awareness

Recycling Recommendations

- Provides disposal instructions
- Suggests recycling best practices
- Encourages responsible waste management

Analytics Dashboard

- Tracks waste detection history
- Visualizes trends using charts
- Supports sustainability monitoring

Sustainability Reports

- Generates automated sustainability summaries
- Highlights waste classification statistics
- Useful for reporting and analysis

RESULTS & OUTPUT

- ✓ Waste classification using MobileNetV2
- ✓ Automatic waste category mapping
- ✓ Environmental impact calculation
- ✓ Recycling recommendation generation
- ✓ Detection history tracking
- ✓ Sustainability report generation

METRIC	RESULT
AI Model	MobileNetV2
Waste Categories	Plastic, Organic, E-Waste
Prediction Confidence	~92% Confidence
Sustainability Score	85/100
Sample CO ₂ Reduction	0.5 kg
Data Storage	CSV History

RESPONSIBLE AI CONSIDERATIONS

- **Fairness:** Model is designed to classify waste categories without bias.
- **Transparency:** Users can see detected objects and waste category predictions.
- **Ethics:** Recommendations promote environmentally responsible disposal practices.
- **Privacy:** Uploaded images are used only for classification and are not linked to personal data.

FUTURE ENHANCEMENTS

Mobile Application

- Develop Android/iOS version for wider accessibility.

Cloud Database Integration

- Replace CSV storage with cloud-based databases.

Advanced AI Models

- Improve classification accuracy using larger datasets.

Real-Time Camera Detection

- Enable live waste recognition through device cameras.

Smart Recycling Center Locator

- Show nearby recycling and disposal centers.

CONCLUSION

- Developed an AI-powered waste classification system using MobileNetV2.
- Automated waste category mapping and recycling recommendations.
- Provided environmental impact analysis and sustainability insights.
- Implemented analytics dashboard and detection history tracking.
- Generated sustainability reports for monitoring waste management activities.
- Contributes to Sustainable Development Goals (SDG 11, SDG 12, and SDG 13).

THANK YOU



Prepared By:

Jeeva A

github.com/Jeeva-0805/RecycleSense-AI

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